

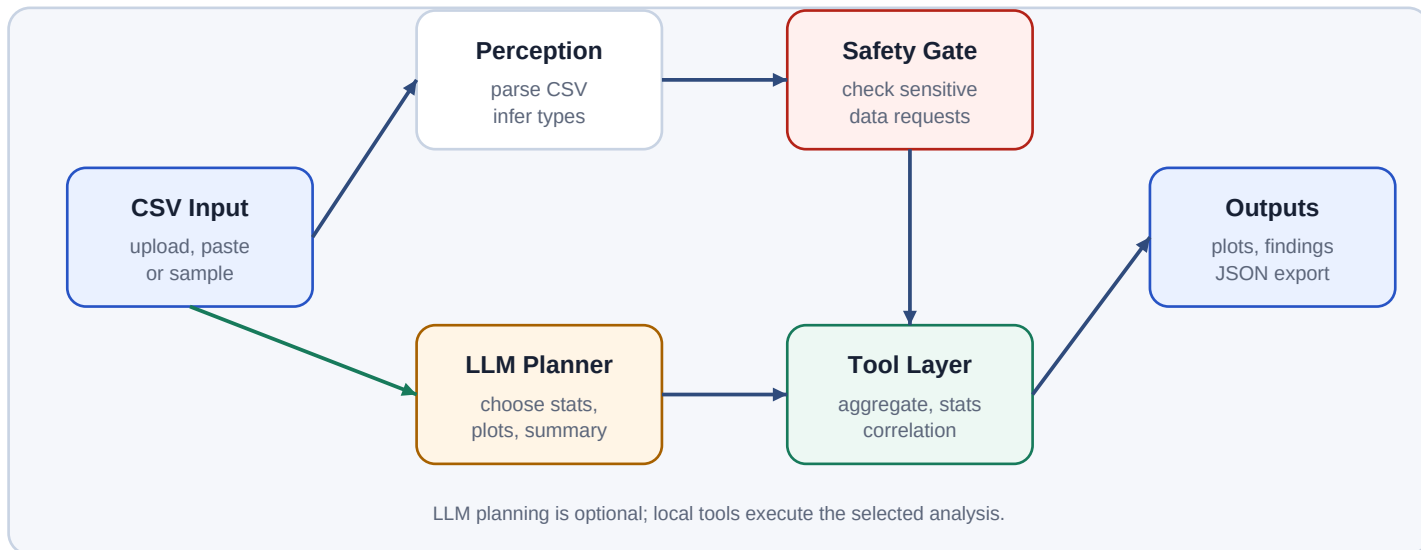
DataLens Agent Report

CSV analysis agent that chooses steps, creates plots, and summarizes findings

GitHub repository: <https://github.com/your-username/datalens-agent> (replace with the uploaded repository URL before submission)

Prototype: A browser-based data analysis agent that accepts pasted or uploaded CSV data and runs an end-to-end analysis loop.

System Design



Design Explanation

DataLens follows a perceive-decide-act agent loop. The user provides a CSV and an analysis goal. The perception module parses the CSV, infers numeric, categorical, and date columns, and checks missing values. The safety gate blocks requests that appear to involve secrets or sensitive identifiers. The optional LLM planner selects suitable analysis steps from the dataset shape, user goal, and checked options; if it is unavailable, deterministic rules select local tools. The tool layer executes schema inspection, numeric profiling, category aggregation, relationship plotting, missing-value checks, outlier scans, trend checks, custom notes, and summary generation.

Agent Concepts Used

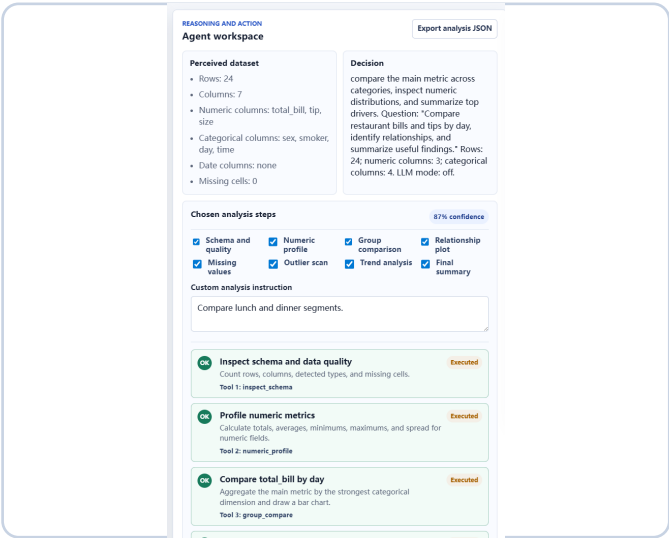
- **Perception:** parses CSV rows and infers column types.
- **Decision making:** optionally uses an LLM planner, with deterministic fallback, to choose or refine selected analysis steps.
- **Action:** executes local analysis tools, draws charts on canvas, handles optional custom instructions, and can use the LLM to rewrite the final summary.
- **Memory:** records loaded datasets, LLM/fallback status, and executed tool runs in localStorage.
- **Safety:** refuses analysis requests involving secrets or sensitive identifiers.

Assignment 2

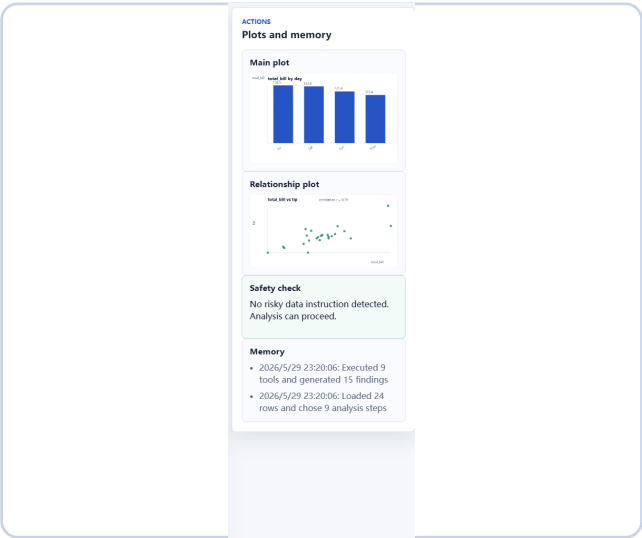
Screenshots and System Behavior

Evidence of CSV perception, step choice, plotting, execution, and summary

The screenshots below show the DataLens agent running a small public tips CSV sample. The agent combines detected fields with selected analysis options, then executes each step.



1. Perception and decisions: the agent detects 24 rows with numeric and categorical fields, then chooses analysis steps.



2. Plot actions: the agent creates a category comparison bar chart and a relationship scatter plot with correlation.

1. inspect_schema: inspect schema and data quality

Detected 24 rows, 7 columns, and 0 missing cells.

2. numeric_profile: Profile numeric metrics

Profiled 3 numeric columns(s).

3. group_compare: Compare total bill by day

Aggregated total bill by day and generated the main bar chart.

4. relationship_plot: Plot relationship between total_bill and tip

Created a relationship plot for total_bill and tip.

5. missing_quality: Check missing values by column

Checked missing values across 7 column(s).

6. outlier_scan: Scan numeric outliers

Scanned 3 numeric column(s) for IQR outliers.

7. trend_analysis: Analyze metric trend over time

Trend analysis skipped because a date column and numeric metric were not both available.

8. custom_note: Apply custom instruction

Recorded and applied the custom analysis instruction.

9. summarize_findings: Summarize findings

Generated final natural-language findings.

Dataset structure: total_bill: numeric, tip: numeric, sex: categorical, smoker: categorical, day: categorical, time: categorical, size: numeric.

Missing cells: 0.

total_bill: total 505.64, average 21.07, range 5.75-40.17

tip: total 78.69, average 3.26, range 1-7.58

size: total 60, average 2.5, range 1-4

Fri leads day with 136.45 total total_bill.

Thur has the lowest total total_bill at 113.42.

total_bill and tip have a correlation of 0.79 in this dataset.

Missing value check: no missing cells were found.

total_bill: 3 possible outlier(s), from 5.75 to 40.17.

tip: 3 possible outlier(s), from 1 to 7.58.

Trend analysis needs at least one date column and one numeric metric.

Custom instruction considered: Compare lunch and dinner segments.

Recommended action: focus first on Fri, because it contributes the strongest total_bill result.

Data quality note: the sample used by the agent has no missing cells.

The agent combined selected user steps with detected column types, then executed tools to produce plots and a summary.

3. Summary: findings explain top categories, numeric profiles, correlation, and recommended business action.

How It Works

- **Input:** user uploads, pastes, or loads a sample CSV.
- **Perception:** CSV parser builds records and infers data types.
- **Decision:** the agent uses checked options and detected fields to choose profiling, aggregation, plotting, quality, custom, and summary steps.
- **Execution:** each selected tool runs and writes to the execution log.
- **Output:** charts, findings, memory, and JSON export make the result reproducible.